

ES denominator defects inside a three-coordinate monodromy system

For a full-shell divisor candidate $u \mid a^2$, $R = 4a - p$, its least return denominator is

$$D_E = \frac{R}{\gcd(R, 4u + 1)}, \quad D_M = \frac{R}{\gcd(R, u + a)}.$$

Integral success at the original prime is exactly $D = 1$.

From the actual $(3, 4, \infty)$ period matrices, act on $(\mathbb{Z}/D)^3$ by

$$\begin{aligned} a_1(x, y, z) &= (6 + y, -6 - x - y, z - 2 + x), \\ a_2(x, y, z) &= (-y, x - 6, z + 3 + y), \\ a_\infty(x, y, z) &= (x, y + x, z - 1). \end{aligned}$$

Every map is invertible. Every cusp cycle has length exactly D . The affine gluing class has exact order D , despite being trivial on each of the two finite cyclic subgroups.

Complete classification for every odd D

	orbit degrees	compact genera
$3 \nmid D$	D^3	$\frac{5D^3 - 12D^2 - 17D + 24}{24}$
$3 \mid D$	$D^3/9$	$\frac{5D^3 - 12D^2 - 81D + 216}{216}$
	$8D^3/9$	$\frac{5D^3 - 12D^2 - 9D + 27}{27}$

At $D = 3$: degrees 3, 24 and genera 0, 1. At $D = 5$: degree 125 and genus 11.

The full labelled union has a section exactly when ES holds. Labels retain every shell, exponent budget, channel, order and clearing scale.